Lecture 7: Trigonometric Substitutions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Perform the integration by using a suitable trigonometric substitutions.

Introduction

Many integrals can be evaluated by using a right substitution, however, there is no general way in finding the right one unfortunately.

Integrals involving the square roots of quadratics often yield to trigonometric substitution.

For $a > 0$,

Expression in integrand	Trigonometric substitution
$\sqrt{a^2 - x^2}$	$x = a \sin\theta$
$\sqrt{a^2 + x^2}$	$x = a \tan\theta$
$\sqrt{x^2 - a^2}$	$x = a \sec\theta$

Alternatively, hyperbolic substitution.

$x = a \tanh\theta$

$x = a \sinh\theta$

$x = a \cosh\theta$.

If you substitute $x = a\sin\theta$ into $\sqrt{a^2 - x^2}$

$$\sqrt{a^2 - a^2\sin^2\theta} = \sqrt{a^2(1-\sin^2\theta)}$$

$$= \sqrt{a^2\cos^2\theta}$$

$$= \sqrt{(a\cos\theta)^2}$$

$$= |a\cos\theta|$$

$\sqrt{x^2} = |x|$



Trigonometric Substitutions

Substitution of $x = a\sin\theta$ in $\sqrt{a^2 - x^2}$ $\quad -\pi/2 \le \theta \le \pi/2$

$dx = a\cos\theta\, d\theta$

Ex: $\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta} = |a\cos\theta|$

$= a\cos\theta$

Substitution of $x = a\tan\theta$ in $\sqrt{a^2 + x^2}$, $\quad -\pi/2 < \theta < \pi/2$

$dx = a\sec^2\theta$

$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2\tan^2\theta} = \sqrt{a^2(1 + \tan^2\theta)}$

$= \sqrt{a^2\sec^2\theta} = |\underset{>0}{a}\,\underset{>0}{\sec\theta}|$

$= a\sec\theta$

Substitution of $x = a\sec\theta$ in $\sqrt{x^2 - a^2}$

$dx = a\sec\theta\tan\theta\, d\theta$

$0 \le \theta < \pi/2$ ✓ $\quad \pi/2 < \theta \le \pi$

$\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2} = \sqrt{a^2(\sec^2\theta - 1)}$

$= \sqrt{a^2\tan^2\theta} = |a\tan\theta| = a\tan\theta.$

When we use the substitution method in the integration, after the integration, we want to "reverse" the substitution.

$g(\theta) \to f(x)$

So we need the substitution to be reversable (invertible)

Ex: $x = a\sin\theta$

$\Rightarrow \theta = \arcsin\left(\frac{x}{a}\right)$

so $-\pi/2 \le \theta \le \pi/2$



For $\sqrt{a^2-(bx)^2}$, substitute $x=\frac{a}{b}\sin\theta$ $dx=\frac{a}{b}\cos\theta d\theta$

$$\sqrt{a^2-(bx)^2} = \sqrt{a^2-\left[b\left(\frac{a}{b}\sin\theta\right)\right]^2}$$

$$= \sqrt{a^2-a^2\sin^2\theta} = a\cos\theta.$$

For $\sqrt{a^2+(bx)^2}$, substitute $x=\frac{a}{b}\tan\theta$

For $\sqrt{(bx)^2-a^2}$, substitute $x=\frac{a}{b}\sec\theta$.

Examples

$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2\sec^2\theta\, d\theta}{\sqrt{4+4\tan^2\theta}}$$

a^2

$a=2$.

Let $x = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

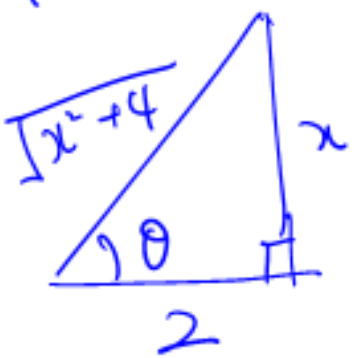
$$= \int \frac{2\sec^2\theta\, d\theta}{\sqrt{4(1+\tan^2\theta)}}$$

Lec 4.

$$= \ln|\sec\theta + \tan\theta| + C$$

$$= \ln\left|\frac{\sqrt{x^2+4}}{2} + \frac{x}{2}\right| + C$$

$\tan\theta = \frac{x}{2}$



$$= \int \frac{2\sec^2\theta\, d\theta}{\sqrt{4\sec^2\theta}}$$

$$= \int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta$$

Examples

$$\int \frac{x^2\,dx}{\sqrt{9-x^2}}$$

a^2

$a = 3$

$$= \int \frac{(3\sin\theta)^2 (3\cos\theta)\,d\theta}{\sqrt{9-(3\sin\theta)^2}}$$

Let $x = 3\sin\theta$

$dx = 3\cos\theta\,d\theta$

$-\pi/2 \le \theta \le \pi/2$

$$= \int \frac{(9\sin^2\theta)(3\cos\theta)\,d\theta}{\sqrt{9-9\sin^2\theta}}$$

$$= \int \frac{(9\sin^2\theta)(3\cos\theta)\,d\theta}{\sqrt{9(1-\sin^2\theta)}}$$

$$= \int \frac{(9\sin^2\theta)(3\cos\theta)\,d\theta}{3\cos\theta}$$

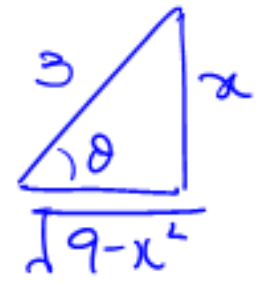
$$= 9\int \sin^2\theta \, d\theta$$

↓ Lec 6.

$$= 9\int \frac{1-\cos 2\theta}{2} \, d\theta$$

$$\sin\theta = \frac{x}{3}$$

$$= \frac{9}{2}\int 1-\cos 2\theta \, d\theta$$



$$= \frac{9}{2}\left[\theta - \frac{\sin 2\theta}{2}\right] + C$$

$$= \frac{9}{2}\left[\arcsin\left(\frac{x}{3}\right) - \frac{2\sin\theta\cos\theta}{2}\right] + C$$

$$= \frac{9}{2}\left[\arcsin\left(\frac{x}{3}\right) - \left(\frac{x}{3}\right)\left(\frac{\sqrt{9-x^2}}{3}\right)\right] + C$$

$$= \frac{9}{2}\arcsin\left(\frac{x}{3}\right) - \frac{x\sqrt{9-x^2}}{2} + C$$

Examples

$$\int \frac{dx}{\sqrt{25x^2-4}},\ x>\frac{2}{5}$$

$bx = 5x$ $\quad a=2$

Let $x = \frac{2}{5}\sec\theta$

$dx = \frac{2}{5}\sec\theta\tan\theta\, d\theta$

$0 \le \theta < \pi/2$

$$= \int \frac{\frac{2}{5}\sec\theta\tan\theta\, d\theta}{\sqrt{25\left(\frac{2}{5}\sec\theta\right)^2-4}}$$

$$= \frac{2}{5}\int \frac{\sec\theta\tan\theta\, d\theta}{\sqrt{4\sec^2\theta-4}}$$

$$= \frac{2}{5}\int \frac{\sec\theta\tan\theta\, d\theta}{\sqrt{4(\sec^2\theta-1)}}$$

$$= \frac{2}{5}\int \frac{\sec\theta \tan\theta \, d\theta}{2\tan\theta}$$

$$= \frac{1}{5}\int \sec\theta \, d\theta$$

$$= \frac{1}{5}\ln\left|\sec\theta + \tan\theta\right| + C$$

$$= \frac{1}{5}\ln\left|\frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2}\right| + C$$

$$\sec\theta = \frac{5x}{2}$$



Examples

$$\int \frac{dx}{(4+x^2)^{\frac{3}{2}}}$$

Exercise.

Ans: $\frac{x}{4\sqrt{x^2+4}} + C$

Examples

$$\int_0^3 \frac{75}{(25-x^2)^{\frac{3}{2}}}\,dx = \int_0^3 \frac{75\ dx}{\left(\sqrt{25-x^2}\right)^3}$$

$a=5$

Let $x = 5\sin\theta$

$dx = 5\cos\theta\, d\theta$

$x=0,\ 5\sin\theta = 0$

$\Rightarrow \theta = 0$

$$= \int_0^{\arcsin(3/5)} \frac{75\,(5\cos\theta)\,d\theta}{\left(\sqrt{25-25\sin^2\theta}\right)^3}$$

$x=3,\ \theta = \arcsin(3/5)$

$-\pi/2 \le \theta \le \pi/2$

$$= \int_0^{\arcsin(3/5)} \frac{75\,(5\cos\theta)\,d\theta}{\left(\sqrt{25(1-\sin^2\theta)}\right)^3}$$

$\cos^2\theta$

$$= \frac{\cancel{25}\times 3}{\cancel{125}\times\cancel{8}} \int_0^{\arcsin(3/5)} \frac{\cos\theta \; d\theta}{\cos^3\theta}$$

$$= 3\int_0^{\arcsin(3/5)} \sec^2\theta \; d\theta$$

$$= 3\left[\tan\theta\right]_0^{\arcsin(3/5)}$$

$$= 3\tan\left[\arcsin(3/5)\right]$$

φ

$$= 3\left(\frac{3}{4}\right) = \frac{9}{4}$$

$\sin\theta = 3/5$

